Dr. Azad Singh

Curriculum Vitae

Department of CSE
Indian Institute of Technology Jodhpur
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My Webpage
Github *Linkedin* *Google Scholar*

Education

2019–2026 **PhD, Computer Science & Engineering, Indian Institute of Technology, Jodhpur**, Self-supervised Encoding of Chest Radiographs for Learning Generalized Representations .

2014–2016 **M.Tech, Computer Science & Engineering, Kurukshetra University**, Digital Forensics, Memory Forensics Percentage: 71.8%.

2009–2013 **B.Tech, Computer Science & Engineering, Kurukshetra University,** Percentage: 70.0%.

2009–2009 **12th, Non-Medical, Aggarsain Public School Kurukshetra**, *CBSE*, Percentage: 64.0%.

Research Interests

Computer Vision, Representation Learning, Self-Supervised Learning, Medical Image Analysis, Vision-Language Modeling, Multimodal Learning.

Technical Skills

Programming Languages: Python (advanced), C/C++

Deep Learning Frameworks: PyTorch, TensorFlow, Keras

Libraries & Tools: OpenCV, Scikit-learn, Git, NumPy, Pandas

Domain Expertise: Computer Vision, Representation Learning, Self-Supervised Learning, Multimodal Learning, Medical Image Analysis

Publications

Journal Articles

2025 **Azad Singh** and Deepak Mishra. Self-supervised contextual representations of chest x-ray images. *IEEE Signal Processing Letters*, 2025. **IF: 3.9**.

2025 **Azad Singh**, Vandan Gorade, and Deepak Mishra. Large scale time-series representation learning via simultaneous low- and high-frequency feature bootstrapping. *IEEE Transactions on Neural Networks and Learning Systems*, 2025. **IF: 10.2**.

2024 **Azad Singh**, Vandan Gorade, and Deepak Mishra. Mlvicx: Multi-level variance-covariance exploration for chest x-ray self-supervised representation learning. *IEEE Journal of Biomedical and Health Informatics*, 2024. **IF: 7.7**.

Communicated Journal Articles

2024 **Azad Singh**, and Deepak Mishra, SiMEX: Siamese Ensembling for Multi-Latent Exploration in Self-Supervised Chest X-ray Representation Learning, In *Computers in Biology and Medicine, Elsevier* **IF:7.7**.

2025 **Azad Singh**, and Deepak Mishra, DiSSECT: Structuring Transfer-Ready Medical Image Representations through Discrete Self-Supervision, In *IEEE Journal of Biomedical and Health Informatics (JBHI)* **IF:7.7** .

2025 Aiman Farooq, ***Azad Singh***, *Deepak Mishra and Santanu Chaudhury*, RobSurv: Multi-Modal Learning for Robust Cancer Survival Prediction, In *Computer Methods and Programs in Biomedicine***IF:4.8** .

In Conference Proceedings and Workshops

2025 **Azad Singh** and Deepak Mishra. Scale-aware adaptive feature quantization for robust medical image representation learning. In ***ECAI***, 2025. CORE **A**.

2025 **Azad Singh**, Wilfred Kisku, Amandeep Kaur, and Deepak Mishra. Vit coupled efficient cmos image sensor with sparse acquisition and patch selection. In ***ISCAS***, 2025. Qualis **A1**.

2025 **Azad Singh**, Vandan Gorade, and Deepak Mishra. Otcxr: Rethinking self-supervised alignment using optimal transport for chest x-ray analysis. In ***WACV***, 2025. CORE **A**.

2025 Gyandera Singh, **Azad Singh**, Aiman Farooq, and Deepak Mishra. Ribcageimp: A deep learning framework for 3d ribcage implant generation. In ***ICVGIP.*** , 2025.

2025 Gajendra Singh, Aiman Farooq, **Azad Singh**, Deepak Mishra, Rahul Chaudhury, and Pushpinder Singh Khera. Guide2heart: Proximity guidance for standard echocardiographic view. In ***Simplifying Medical Ultrasound, ASMUS, MICCAI Workshop***, 2025. CORE **A**.

2024 **Azad Singh** and Deepak Mishra. Idqce: Instance discrimination learning through quantized contextual embeddings for medical images. In ***ICPR***, 2024. CORE **B**.

2024 **Azad Singh** and Deepak Mishra. Coboom: Codebook guided bootstrapping for medical image representation learning. In ***MICCAI***, 2024. CORE **A**.

2022 **Azad Singh** and Deepak Mishra. Branching out for better byol. In *NeurIPS Workshop: Self-Supervised Learning - Theory*, 2022. CORE **A***.

2022 Vandan Gorade, **Azad Singh**, and Deepak Mishra. Mbgrlp: Multiscale bootstrap graph representation learning on pointcloud. In *Student Abstract, AAAI*, 2022. CORE **A***.

Fellowships & Awards

2025 **Oral Presentation** – Scale-Aware Adaptive Feature Quantization for Robust Medical Image Representation Learning, *ECAI 2025*, Bologna, Italy, October 27–30, 2025

2025 **Poster Presentation** – Graduate Forum, *IndoML 2025*, BITS Pilani – Hyderabad Campus, December 2025

2025 **Invited Presentation** – *ConfAI 2025*, Plaksha University, Mohali, November 28–30, 2025 (Accommodation, Registration, and Travel Supported)

2025 **Best Poster (Runner-Up)** – Guide2Heart: Proximity Guidance for Standard Echocardiographic View, *ASMUS Workshop, MICCAI 2025*, Daejeon, Republic of Korea, September 27, 2025

2024 ACM India–IARCS Travel Grant

2024 RISE–MICCAI & Anusandhan National Research Foundation (ANRF) Travel Support

2021 Best Teaching Assistant Award – Data Structure & Algorithms

2019–2023 MHRD Scholarship – Doctoral Fellowship for PhD Research

Experience

Feb 2026 – Present PhD Researach Intern (LLM Driven 3D Medical Segmentation), Microsoft Reserach, Banglore, India

Aug 2025 – Feb 2026 Assistant Professor, School of Computer Science & Engineering, Bennett University, Greater Noida, India

2017-2019 Assistant Prof. (Contract), National Institute of Technology, Hamirpur, H.P, India, 2 years

2017-2017 Assistant Prof., JMIT Radaur, Haryana, 4 Months

Referees

Dr. Deepak Mishra
Asst. Professor, Dept. of CSE
Indian Institute of Technology, Jodhpur
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Dr. Gaurav Gupta
Scientist 'F', Ministry of Electronics and IT
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